

Hybrid Renewable Energy Generation Solution

Mini Project - Daily Progress Report (June 1 - June 13)

This document records the daily progress of the project from the planning and requirement-gathering phase through database and UI design, covering Day 1 (June 1) to Day 13 (June 13).

Day 1 – June 1, 2026

Planned Activity: Project Title Finalization

Expected Deliverable: Approved Project Title

Explanation:

On the first day, the project topic 'Hybrid Renewable Energy Generation Solution' was proposed and finalized. The team conducted a brief discussion on the relevance of combining multiple renewable energy sources such as solar and wind to ensure consistent and reliable power generation. The title was reviewed against project guidelines and approved as the official title for the mini project.

Day 2 – June 2, 2026

Planned Activity: Requirement Gathering

Expected Deliverable: Problem Statement

Explanation:

The second day focused on identifying the core problem the project aims to solve, namely the inconsistency and unreliability of standalone renewable energy sources like solar (dependent on sunlight) and wind (dependent on wind speed). Existing systems and research papers related to hybrid renewable energy systems were studied to understand current limitations. Based on this study, a clear problem statement was drafted, emphasizing the need for a hybrid system that intelligently combines multiple energy sources to maximize efficiency and minimize power loss.

Day 3 – June 3, 2026

Planned Activity: User & Module Identification

Expected Deliverable: Project Objectives

Explanation:

On this day, the different types of users who would interact with the system were identified, including administrators, energy monitoring operators, and end users who consume the generated power. The major functional modules of the application were also identified, such as energy source management, real-time monitoring, data analytics, and reporting. Based on these, the overall project objectives were defined, focusing on building a system that monitors, manages, and optimizes the combined output of solar and wind energy sources.

Day 4 – June 4, 2026

Planned Activity: Use Case Diagram Preparation

Expected Deliverable: Module List

Explanation:

The fourth day was dedicated to preparing a Use Case Diagram that illustrates the interactions between the identified users (actors) and the system's functionalities. Use cases such as 'Login', 'View Energy

Generation Data', 'Manage Energy Sources', 'Generate Reports', and 'Monitor System Status' were mapped out. From this exercise, a finalized module list was prepared, consolidating all the functional units required for the Hybrid Renewable Energy Generation Solution.

[Insert Use Case / UML Diagram for June 4 here]

Day 5 – June 5, 2026

Planned Activity: Database Requirement Analysis

Expected Deliverable: CRUD APIs

Explanation:

On the fifth day, the data requirements for the system were analyzed in detail. Entities such as Energy Source, Energy Generation Record, User, and Report were identified along with their attributes. Based on this analysis, the Create, Read, Update, and Delete (CRUD) operations required for each entity were listed out, forming the basis for the backend API design that would later be implemented using Spring Boot.

Day 6 – June 6, 2026

Planned Activity: ER Diagram Design

Expected Deliverable: Table List

Explanation:

The sixth day focused on translating the identified entities and their relationships into an Entity-Relationship (ER) Diagram. Relationships such as a User monitoring multiple Energy Sources, and an Energy Source generating multiple Energy Generation Records over time, were modeled. From this ER Diagram, the final list of database tables was derived, including tables for Users, Energy Sources, Energy Records, and System Reports.

[Insert ER Diagram for June 6 here]

Day 7 – June 7, 2026

Planned Activity: Database Schema Creation

Expected Deliverable: ER Diagram

Explanation:

On this day, the ER Diagram designed previously was refined and finalized, incorporating primary keys, foreign keys, and constraints for each table. The relationships between the Users, Energy Sources, Energy Generation Records, and Reports tables were clearly defined, ensuring referential integrity. This finalized ER Diagram served as the blueprint for the actual database schema creation.

[Insert Finalized ER Diagram for June 7 here]

Day 8 – June 8, 2026

Planned Activity: UI Wireframe Design

Expected Deliverable: SQL Schema

Explanation:

The eighth day involved creating low-fidelity wireframes for the major screens of the application, including the login page, dashboard, and data entry forms for energy source details. Simultaneously, the SQL schema was written based on the finalized ER Diagram, defining all tables, columns, data types, and relationships required to support the Hybrid Renewable Energy Generation Solution database.

Day 9 – June 9, 2026

Planned Activity: Login & Dashboard UI Design

Expected Deliverable: Page Layouts

Explanation:

On this day, the user interface design was extended to cover the Login page and the main Dashboard page in detail. The Dashboard was designed to display key metrics such as current energy output from solar and wind sources, total energy generated, and system status indicators. Page layouts for these two critical screens were finalized, ensuring a clean and user-friendly design suitable for monitoring renewable energy data.

Day 10 – June 10, 2026

Planned Activity: Navigation & Form Design

Expected Deliverable: UI Screens

Explanation:

The tenth day focused on designing the navigation structure of the application, including the side menu and top navigation bar, to allow easy movement between modules such as Dashboard, Energy Sources, Reports, and Settings. Additionally, form designs for adding and updating energy source details (such as source type, capacity, and location) were created. These efforts resulted in a complete set of UI screens for the application.

Day 11 – June 11, 2026

Planned Activity: Design Review

Expected Deliverable: UI Prototype

Explanation:

On this day, all the UI designs created so far - including the login page, dashboard, navigation, and forms - were reviewed for consistency in color scheme, typography, and layout. Feedback was incorporated to improve usability, particularly for the energy monitoring dashboard which displays real-time data from solar and wind sources. The result of this review was a polished UI Prototype ready

for stakeholder approval.

Day 12 – June 12, 2026

Planned Activity: Frontend Environment Setup

Expected Deliverable: Design Approval

Explanation:

The twelfth day involved setting up the frontend development environment, including installation of Node.js, npm, and the necessary frontend framework dependencies. The project folder structure was initialized in preparation for development. Alongside this setup, the finalized UI prototype was presented and received Design Approval, marking the completion of the design phase and the transition into the development phase.

Day 13 – June 13, 2026

Planned Activity: React Project Setup

Expected Deliverable: React Project Setup

Explanation:

On the thirteenth day, the React project for the frontend of the Hybrid Renewable Energy Generation Solution was initialized. The basic folder structure was created, including directories for components, pages, services, and assets. Essential dependencies such as routing libraries and UI component libraries were installed and configured, setting up a solid foundation for the upcoming development of the Login Page, Dashboard, and other application modules in the following days.